Lesson 2.5 Guided Notes

Session 1 · Range and IQR

I can find the range and the interquartile range of a data set. · 6.SP.B.5c

Word / Spanish	Meaning — complete the blank	Example
range / rango	The greatest value _____ the least value.	20 − 5 = 15
interquartile range (IQR) / rango intercuartil	Q3 minus Q1; the spread of the _____.	18 − 8 = 10
extreme value / valor extremo	A value far from the others; it changes the _____ a lot.	62 in 11, 14, 16, 19, 62

Key idea

Range and IQR both measure _____, or variability.

For this seven-value example, increasing the maximum leaves the _____ unchanged. Recompute quartiles for a different set.

Worked example

Hikes (km): 5, 12, 8, 20, 15, 10, 18. Find the range and the IQR.

Order: 5, 8, 10, 12, 15, 18, 20. Range = 20 − 5 = _____.

Q1 = _____ and Q3 = _____.

Answer: IQR = 18 − 8 = _____ km

You try

1. Data: 4, 9, 6, 15, 11, 7, 13.
a. Range: _____
b. IQR: _____

_____

2. A set has min 12 and range 25. Find the max.

_____

Sum it up: The IQR is often better than the range when ___ because ___.

_____

Session 2 · Comparing Variability

I can compare the variability of two data sets using range and IQR. · 6.SP.B.5c

Word / Spanish	Meaning — complete the blank	Example
consistent / constante	Values are close together; the spread is _____.	IQR 4 vs. IQR 12
measure of variability / medida de variabilidad	A number that tells how _____ data are.	range, IQR
extreme value / valor extremo	A value far from the rest; it stretches the _____.	40 in 0, 3, 10, 18, 40

Key idea

Compare the _____ measure for both groups. The smaller value means more consistent.

When one value is far from the rest, trust the _____ more than the range.

Worked example

Class R: range 18, IQR 12. Class S: range 35, IQR 25. Which class is more consistent?

Ranges: 18 < 35, so R is _____ spread overall.

IQRs: 12 < 25, so R's _____ is closer together.

Answer: Class _____ is more consistent.

You try

1. Find each IQR. Circle the more consistent set.
a. P: Q1 = 30, Q3 = 42 _____
b. Q: Q1 = 33, Q3 = 39 _____

_____

2. Set A: range 50, IQR 8. Set B: range 22, IQR 16. Which set likely has one unusual value?

_____

Sum it up: To decide which group is more consistent, I compare ___ because ___.

_____

Session 1 · Model, reason, and explain

Range and IQR · 6.SP.B.5c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Use the medians of the lower and upper halves. Compare A: 1, 2, 3, 4 and B: 1, 2, 3, 40. Find each range and IQR. Is changing a maximum guaranteed to leave IQR unchanged?

A: 1, 2 | 3, 4

B: 1, 2 | 3, 40

_____

_____

_____

Extend your reasoning

Explain why replacing 7 by 70 in 1, 2, 3, 4, 5, 6, 7 leaves IQR unchanged under this convention.

_____

_____

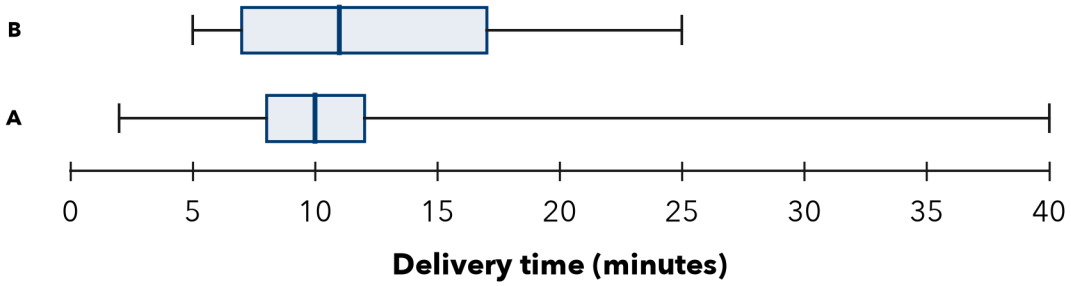
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Comparing Variability · 6.SP.B.5c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Two delivery routes have five-number summaries A: 2, 8, 10, 12, 40 and B: 5, 7, 11, 17, 25 minutes. Compare overall spread and middle-half spread. Which route is more consistent depends on which measure?



_____

_____

_____

Extend your reasoning

Recommend a route for someone who prioritizes a predictable typical trip, then explain what risk your choice does not capture.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?